# ⚡ Spark Plug & Spark Check: Lab Sheet · ANSWER KEY

Small Engines Overhaul · Unit 6 · Level 3 · goes with Ignition Handout

Name: _____

**Question:** Is the engine's ignition system making a good spark, and what does the spark plug show?

## Materials

plug socket, gap gauge, spark tester, gauge card, plug chart

## ⚠️ Safety

- Safety glasses
- Plug wire disconnected
- Use the spark tester; never hold the wire
- Tools with adult supervision

## Prediction

Predict the plug's color and what it means.

**Answers vary.**

## Procedure

1. Remove plug; read color.
2. Measure and set gap.
3. Reinstall plug.
4. Spark test.
5. Check air gap.

## Data

| Check | Result | Spec / normal |
|---|---|---|
| Plug color | **black** | light tan |
| Plug gap | **0.028** | **0.030** |
| Spark color | **blue** | bright blue |
| Air gap | **0.010** | **0.010** |

*Expected results (your class data may differ).*

## Analysis (Claim, Evidence, Reasoning)

1. Claim: is the ignition OK? Evidence: spark and gaps. Reasoning: what the plug color says about the fuel mixture.
**Spark is good and gaps in spec; black plug means rich, so check choke/air filter.**

🔑 *Key written by our team from the class teacher's lab sheets, please check.*